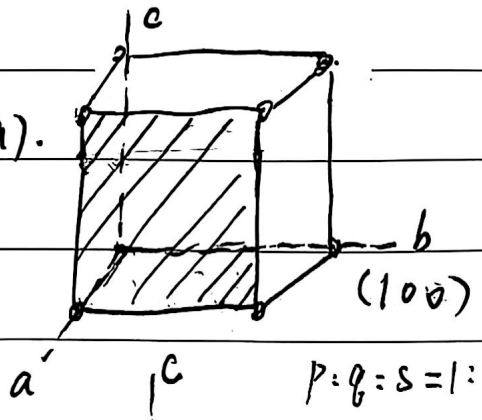
1.1

(a). $6\times\frac{1}{2}+8\times\frac{1}{8}=4$

(b). $1+8\times\frac{1}{8}=2$

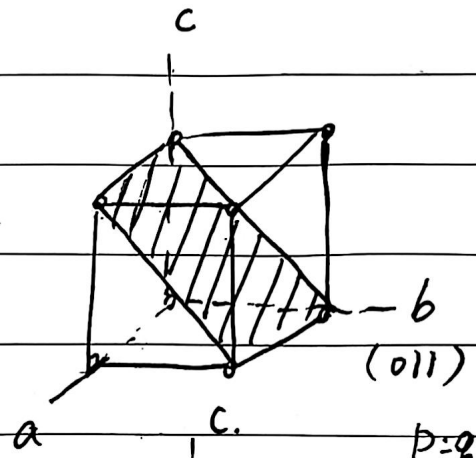
(c) $4+6\times\frac{1}{2}+8\times\frac{1}{8}=8$

1.6 (a).



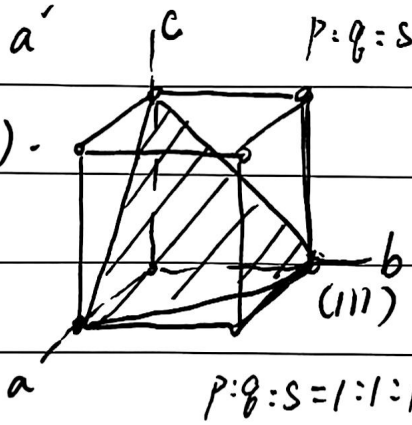
$p:q:s=1:\infty:\infty$

(b).



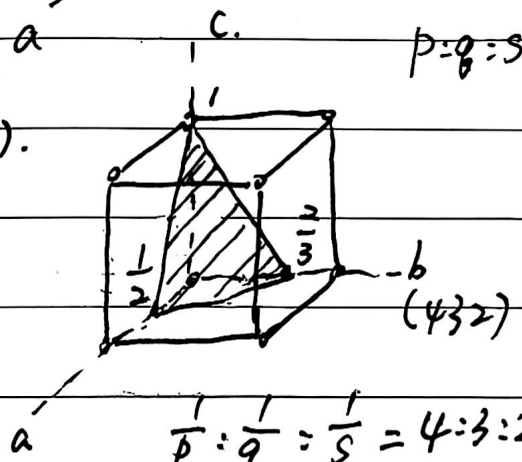
$p:q:s=\infty:1:1$

(c).



$p:q:s=1:1:1$

(d).

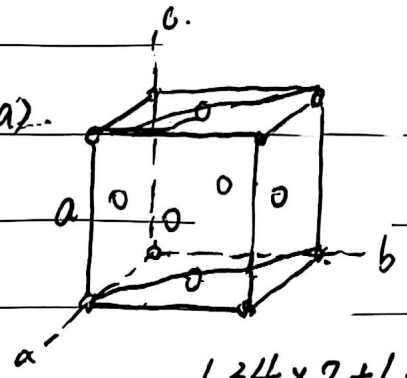


$\frac{1}{p}:\frac{1}{q}:\frac{1}{s}=4:3:2$

则 $p:q:s=3:4:6$

$=\frac{1}{2}:\frac{2}{3}:1$

1.10 (a).



$$a=\frac{1.34\times2+1.34\times2}{\sqrt{2}}=1.34\times2\sqrt{2}\,\text{Å}$$

$$\text{体密度}=\frac{4}{(1.34\times2\sqrt{2}\times10^{-10}\,m)^3}=7.347\times10^{28}\ \text{原子}/m^3$$

$$=7.347\times10^{22}\ \text{原子}/cm^3$$

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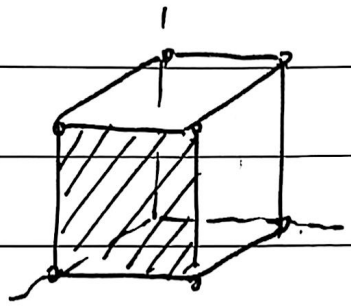
(b). $d = \frac{a}{\sqrt{2}}$

$= \frac{1}{\sqrt{2}} \times 1.34 \times 2\sqrt{2}\,\text{Å} = 2.68 \times 10^{-8}\,\text{cm}$.

(c)



$\frac{1}{p} : \frac{1}{q} : \frac{1}{s} = 1 : 0 : 0$

则 $p : q : s = 1 : \infty : \infty$

面密度 $= \frac{2}{(1.34 \times 2\sqrt{2}\,\text{Å})^2} = 1.39 \times 10^{15}$ 原子/cm^2

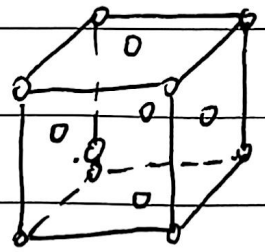
补充题:

1. 设半径为R. $a = \frac{4R}{\sqrt{3}} \times 2 = \frac{8R}{\sqrt{3}}$.

空隙率 $= 1 - \frac{(4+3+1) \times \frac{4}{3}\pi R^3}{a^3}$

$= 1 - \frac{\frac{32}{3}\pi R^3}{\left(\frac{8R}{\sqrt{3}}\right)^3} \times 100\% = 66.0\%$

2.



设一个原子的半径为R, 晶格常数 $a = \frac{4R}{\sqrt{2}}$ cm.

$\rho = \frac{4 \times m}{\left(\frac{4R}{\sqrt{2}}\right)^3}$, $\frac{m}{\frac{1}{12} m_{碳}} = M$.

$m_{碳} = \frac{12g}{6.02 \times 10^{23}}$

$$\frac{4\cdot M\cdot\frac{1}{12}\cdot\frac{12g}{6.02\times10^{23}}}{a^3}=8.94g/cm^3.$$

则 $a=9.06\sqrt[3]{M}\times10^{-9}\,cm$. （晶格常数）.

$d_{min}=2R=\frac{a}{\sqrt{2}}=6.41\sqrt[3]{M}\times10^{-9}cm$.

因为 $\rho=8.94g/cm^3$，故金属为铜，$M=64$

则 $\begin{cases} a=3.62\times10^{-8}cm. \\ d_{min}=\frac{a}{\sqrt{2}}=2.56\times10^{-8}cm. \end{cases}$

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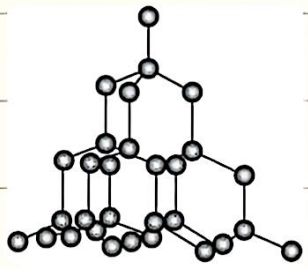
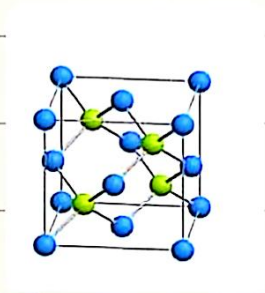
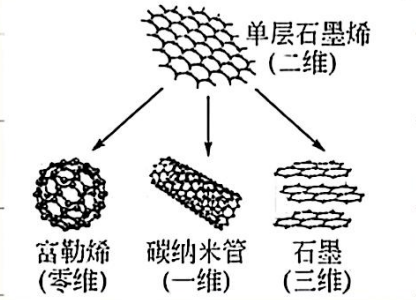
补充题：

元素：① 硅：金刚石结构半导体，硅基材料核心元素。

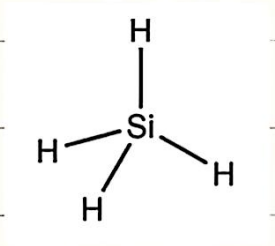


② 碳：构成金刚石等多种晶体，其中金刚石的每个C原子均与相邻的4个C原子形成稳定正四面体结构。

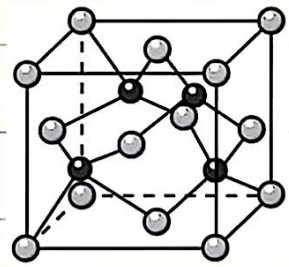




③ 氢：可与Si形成Si-H键，可参与范德华力相关研究。



化合物：① 碳化硅(SiC)：由硅碳原子以强共价键形成，主要为闪锌矿结构，高硬度、耐高温。



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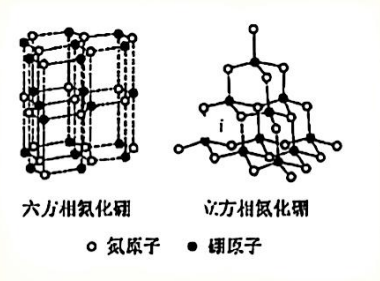
② Ⅲ-Ⅴ化合物：氮化硼（BN）、磷化硼（BP）、氮化铝（AlN）

BN：有六方（类石墨、润滑绝缘）和立方（类金刚石、超硬）结构。

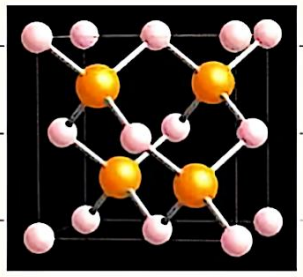
BP：闪锌矿结构，可作高硬度、宽禁带半导体、耐高温。

AlN：高导热、绝缘性好，用于LED散热基板、高频电路等。

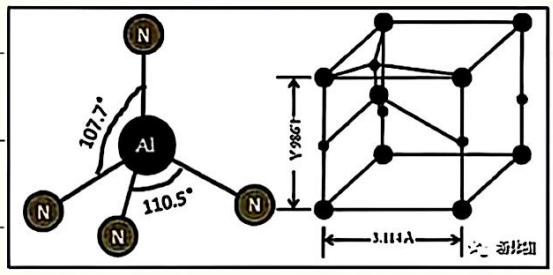
BN



BP



AlN



③ 半导体：硫化铅（PbS）、碲化铅（PbTe）、硒化铅（PbSe），

均为Ⅳ-Ⅵ族化合物，均为窄禁带半导体，应用于光电、能源领域

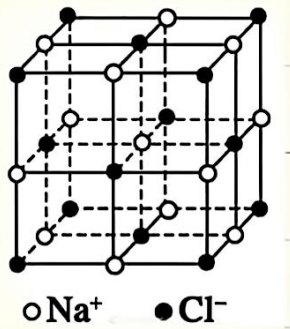
PbS

PbTe

PbSe

④ 离子晶体：氯化钠（NaCl）、氯化钾（KCl）氯化铯（CsCl）

氯化钠：Na^+与Cl^-以离子键结合，易溶于水，食盐。



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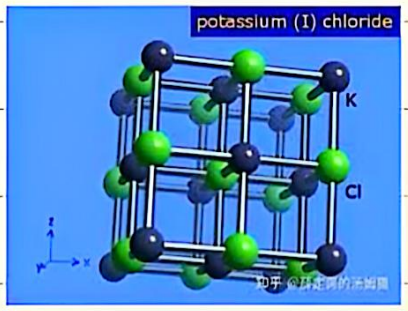
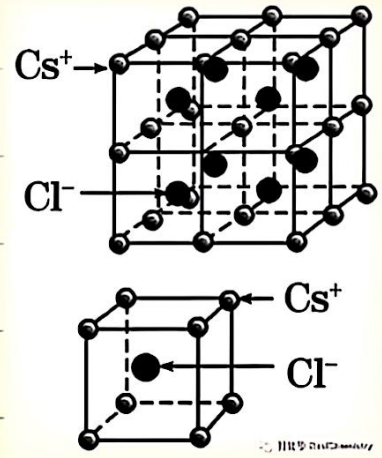
氯化钾：面心立方结构，离子键结合，可作钾肥，电解质等。



氯化铯：体心立方结构，离子键结合，溶解度较高，常用于结构研究模型。